

matlab4jax

Run MATLAB code from JAX with `jax.jit` support.

Example

```
import jax
import jax.numpy as jnp
from jax import ShapeDtypeStruct

import matlab4jax

x = jnp.array([[1.0, 2.0],
               [3.0, 4.0]])

[y] = jax.jit(lambda x: matlab4jax.run_matlab(
    inputs=[x],
    input_names=["x"],
    command="y = inv(x);",
    output_names=["y"],
    abstract_outputs=[ShapeDtypeStruct(x.shape, x.dtype)]
))(x)

print(y)
```

Output:

```
[[ -2.00000002  1.00000001 ]
 [ 1.50000001 -0.50000006]]
```

Installation

Requires:

- JAX
- MATLAB with the MATLAB Engine API
- C++ compiler and CMake

Use `pip install .` to install `matlab4jax`.

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Matlab library path

On Linux, set the Matlab library path before importing, for example:

```
export
```

```
LD_LIBRARY_PATH=/usr/local/MATLAB/R2026a/extern/bin/glnxa64:$LD_LIBRARY_PATH
```

```
H
```

To make this setting persistent, add it to your `~/.bashrc`:

```
echo 'export
```

```
LD_LIBRARY_PATH=/usr/local/MATLAB/R2026a/extern/bin/glnxa64:$LD_LIBRARY_PATH
```

```
H' >> ~/.bashrc
```

```
source ~/.bashrc
```